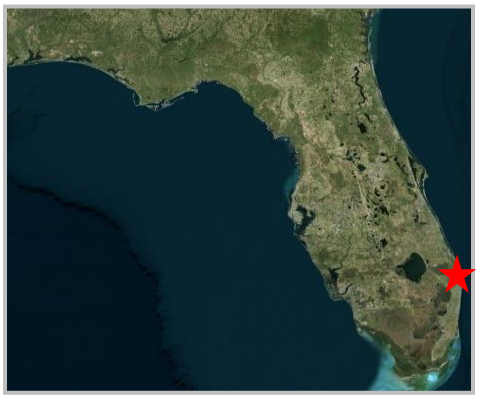
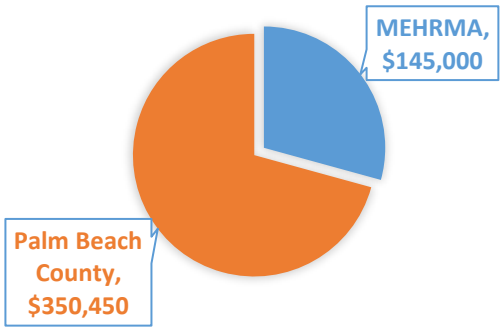
Highland Beach Mangrove Restoration, Palm Beach County, FL





Partner

- Palm Beach County Department of Environmental Resources Management

Project Description

The Highland Beach Mangrove Project consists of 24 acres of estuarine habitat, predominantly mangroves, divided into two parcels separated by a housing community, located in the Town of Highland Beach, in southern Palm Beach County. The mangrove areas are owned by Palm Beach County or are in private ownership and under conservation easements. The narrowness of the ICW and lack of a speed zone in the project area is causing severe erosion to the naturally existing mangroves. In fact, aerial imagery shows that since the 1940's, the shoreline in this area has receded shoreward over 60 feet.

The purpose of this project was to preserve and enhance 17.6 acres of mangroves by installing a 1,700 linear foot (0.8 acre) limestone riprap breakwater. The limestone attenuates waves, reduces the effects of boat wakes, prevents scouring and protects seagrass beds from sedimentation in the project area. The rock serves as substrate for the recruitment and colonization of oysters, sponges, and algae. The enhanced habitat provides nursery and foraging habitat for a variety of wildlife including juvenile fish, invertebrates, oysters, wading birds, manatees and juvenile green sea turtles.

Benefits

- Attenuate waves
- Conserve native mangrove community
- Reduce the effects of boat wakes
- Prevent scouring and protect seagrass beds from sedimentation in the project area
- Provide substrate for the recruitment and colonization of oysters, sponges, and algae
- Provide nursery and foraging habitat for a variety of wildlife including juvenile fish, invertebrates, oysters, wading birds, manatees and juvenile green sea turtles.

FWC Strategic Plan Elements

This project directly addresses the following FWC Strategic Plan Elements:

Theme I, Goal 2: Ensure sufficient habitats exist to support healthy and diverse fish and wildlife populations.

Strategies:

2. *Protect lands and waters critical to sustaining healthy and diverse fish and wildlife populations through diverse incentive programs.*
3. *Manage habitats to sustain healthy and diverse fish and wildlife populations.*

Acreage restored by habitat type (footprint)

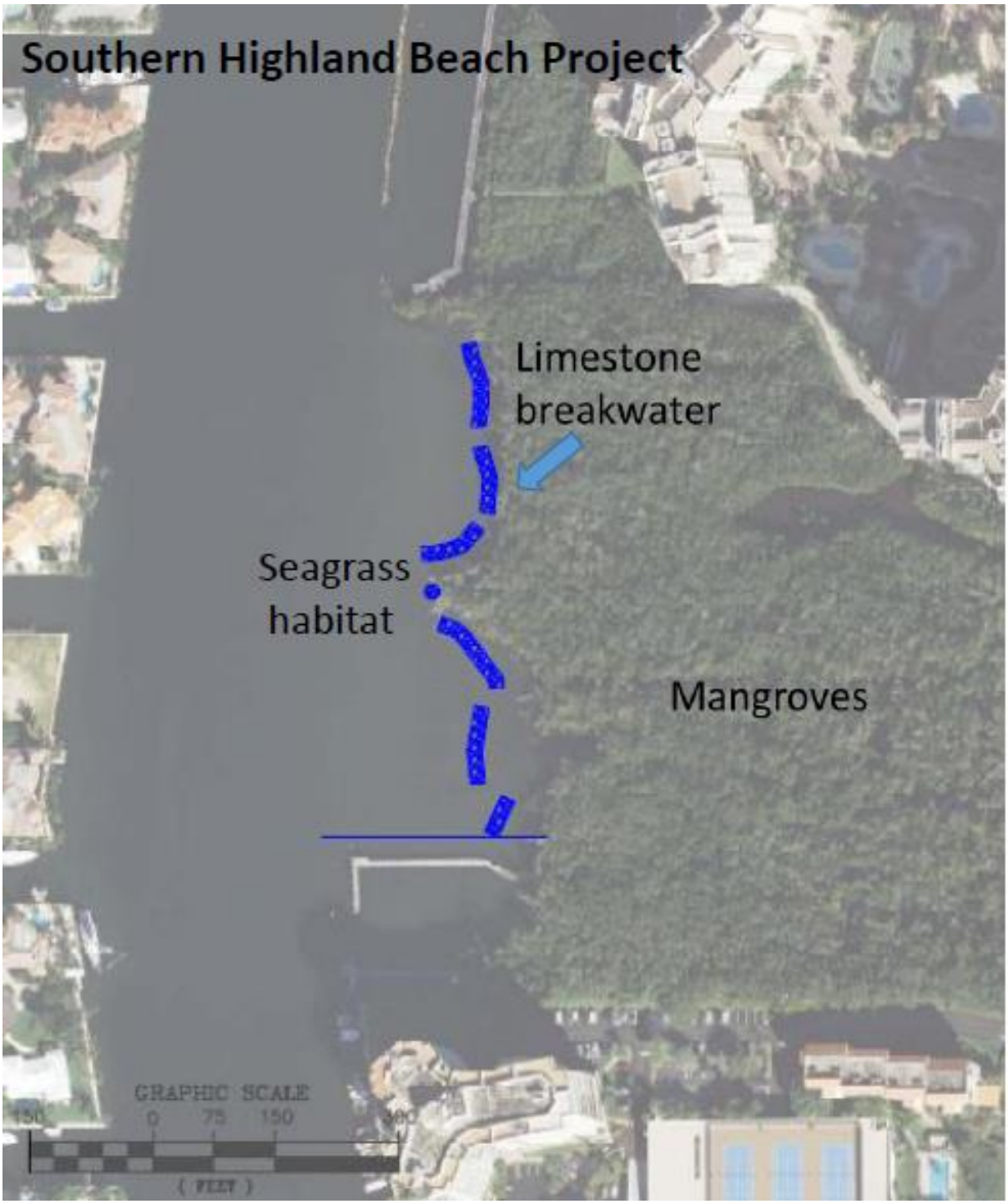
- 1,700 linear feet (0.8 acre) of limestone rip rap breakwater placed to:
  - prevent shoreline erosion and protect 17.6 acres of mangroves
  - enhance 3.2 acres of seagrass habitat by reducing sedimentation from erosion

Acreage affected by restoration activities

- Total project acreage, including unvegetated estuarine bottom is 24 acres

Project Map (identify acreage of both footprint and affected area)
Total project acreage: 0.8 acre limestone breakwater, 17.6 acres mangrove, 3.2 acre seagrass




Southern Highland Beach Project
Limestone breakwater
Seagrass habitat
Mangroves
GRAPHIC SCALE
0 75 150
(FEET)

Project Photos

Highland Beach Aerial Survey
Pre-construction Northern Phase (MEHRMA funded portion of project)

Date: 3/24/2015
Time: 11:00am-12:00pm
Altitude: 70'-350'
Duration: 1hr (two 25minute flights)
GPS Start Point: 26 24 52N, 80 4 3W
GPS End Point: 26 24 22N, 80 4 5W
Photos: North→South

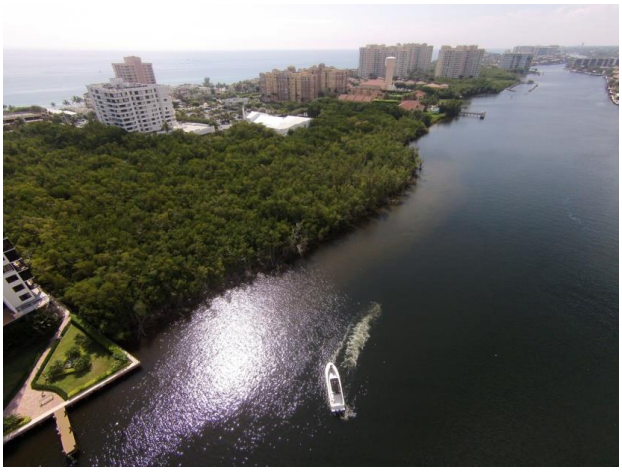


Photo 1: Pre-construction northern phase, facing south

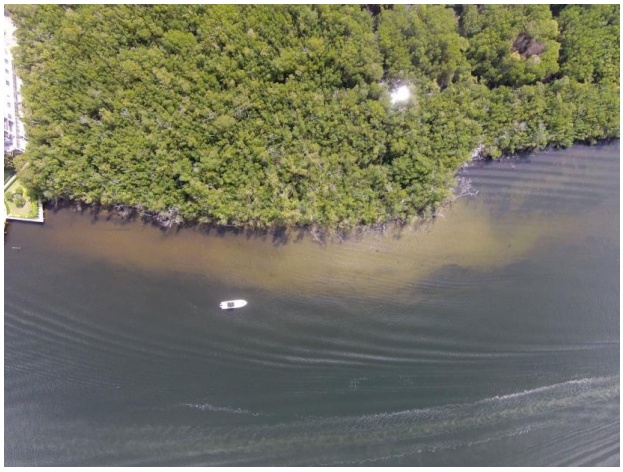


Photo 2: Pre-construction northern phase, facing east



Photo 3: Pre-construction northern phase, facing east



Photo 4: Pre-construction northern phase, facing northeast

Highland Beach Aerial Survey
Post-Construction Northern End (MEHRMA funded portion of project)

Date: 5/1/2015
Time: 1:30pm-2:30pm
Altitude: 50'-400'
Duration: 1hr (two 25minute flights)
GPS Start Point: 26 24 52N, 80 4 4W
GPS End Point: 26 24 20N, 80 4 7W

Photos: North→South



Photo 1: northern phase, facing south



Photo 2: northern phase, facing down

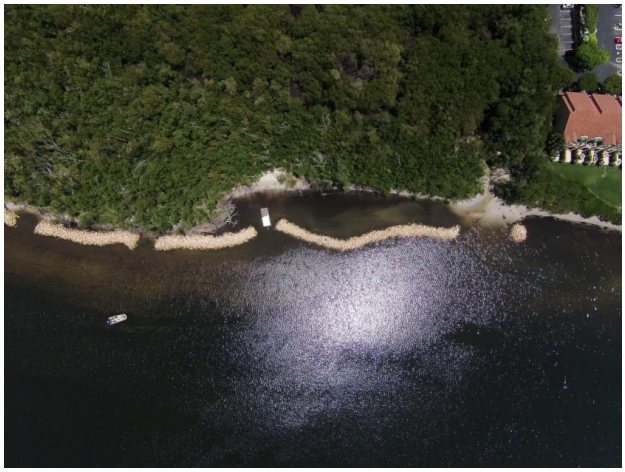


Photo 3: northern phase, facing down



Photo 4: northern phase, facing north